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840 (concluded). On Mascheroni's Geometry of the Compass.

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and by construction, $(EG)^2 = (CF)^2$; that is,

$$(EG)^2 = (CE)^2 + \text{rad.}^2, = (CE)^2 + (CG)^2;$$

viz. CEG is a right-angled triangle; that is, CG is at right angles to BCE , or the line CGF bisects the arc AB in G .

For the second problem, (2), to find the intersection of two given lines, we require the solution of the problem to find a fourth proportional to three given distances, and this immediately depends on the following problem.

Problem. In two concentric circles to place chords subtending equal angles. If AB (fig. 2) be a chord of the one circle, then with centres A and B respectively, and

[Figure 2: Two concentric circles with centre C ; chord AB in the inner circle and chord $A'B'$ in the outer; the points A' , B' obtained by intersecting an arc of radius $AA' = BB'$ centred at A and B respectively with the larger circle.]

an arbitrary radius $AA' = BB'$, describing circles to cut the larger circle in the points A' and B' (A' is either of the two intersections and B' is the intersection lying in regard to CB as A' lies in regard to CA) then clearly $\angle A'CB = \angle ACB'$. And hence also we have the following problem.

Problem. To find a fourth proportional to three given distances. We have then only to take as the given distances the two radii CA , CA' and the chord AB ; and $A'B'$ is a fourth proportional to CA , CA' , AB . From the similar triangles ACB , $A'CB'$, we have

$$CA : CA' :: AB : A'B', \quad \text{viz.}$$

$A'B'$ is a fourth proportional to CA , CA' , AB .

We have now the solution of

(2) To construct the intersection of two given lines AB and CD (fig. 3). Find

[Figure 3: Two lines AB and CD ; their counter-points c of C and d of D with respect to AB , the parallelogram $Cy dD$, and the required intersection point S constructed on two circles of radii cS centred at c and C .]

c the counter-point of C , and d the counter-point of D , in regard to the line AB ;

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and find y such that the distances Cy , dy are $= Dd$, DC respectively; y is in a line with cC , and we have $Cy dD$ a parallelogram. Find cS a fourth proportional to cy , cd , cC , and with centres c , C respectively and radii each cS describe circles cutting in the point S ; this will be the required intersection of the two lines. In fact, the required point S will be the intersection of the two lines CD , cd : supposing these lines each of them drawn, and also the lines cCy and dy , we have DC parallel to dy , that is, the triangles cdy , cSC are similar to each other or $cC : cy :: cd : cS$, viz. the distance cS having been found by this proportion,

and the point S found as the intersection of the two circles, centres c and C respectively, the point S so determined is the required point of intersection of the given lines AB and CD .

If a circle be given without its centre being known, then taking any three points A, B, C on the circle, and a pair of counter-points D, E of the line AB , and also a pair of counter-points F, G of the line AC , we have obviously the centre of the given circle as the intersection of the lines DE and FG ; and the centre can thus be found with the compass only.

It is proper to remark that the problems considered in the present paper are those connecting the theory with ordinary geometry, not the problems which are most readily and elegantly solved with the compass only: a large collection of these are contained in the work, and in particular the twelfth book contains some interesting *approximate* solutions of the problems of the quadrature of the circle, the duplication of the cube, and other problems not solvable by ordinary geometry.

Cambridge, March 19, 1885.

841. On a Differential Operator.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 190, 191.]

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Write $X = 1 + bx + cx^2 + \dots$, $= (1 - \alpha x)(1 - \beta x)(1 - \gamma x) \dots$; then by Capt. MacMahon's theorem, any non-unitary function of the roots $\alpha, \beta, \gamma, \dots$ is reduced to zero by the operation

$$\Delta_x = \partial_b + b\partial_c + c\partial_d + \dots;$$

for instance, if

$$(2)_x = \Sigma \alpha^2 = b^2 - 2c,$$

we have

$$\Delta(b^2 - 2c) = 2b + b(-2), = 0.$$

We have

$$\Delta X = x + bx^2 + cx^3 + \dots = (xX).$$

And writing, moreover, $X' = b + 2cx + 3dx^2 + \&c.$, for the derived function of X , then

$$\Delta X' = 1 + 2bx + 3cx^2 + \dots = (xX)'.$$

We can hence shew that $\Delta\left(\frac{X'}{X} - b\right) = 0$; the value is, in fact,

$$\frac{\Delta X'}{X} - \frac{X' \Delta X}{X^2} - \Delta b, \quad \text{that is,} \quad \frac{(xX)'}{X} - \frac{X' x X}{X^2} - 1,$$

which is

$$= -\frac{X + xX'}{X} - \frac{xX'}{X} - 1.$$

This is right, for $\frac{X'}{X}$ is a sum of non-unitary symmetric functions of the roots; in fact,

$$\frac{X'}{X} = \Sigma \frac{-\alpha}{1 - \alpha x} = -(1) - (2)x - (3)x^2 - \&c.,$$

or since $b = -(1)$, this is

$$\frac{X'}{X} - b = -(2)x - (3)x^2 - \&c.,$$

a sum of non-unitary functions of the roots.

842. On the Value of $\tan(\sin \theta) - \sin(\tan \theta)$.

[From the *Messenger of Mathematics*, vol. XIV. (1885), pp. 191, 192.]

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The following equation is given p. 59 of the *Lady's and Gentleman's Diary* for 1853:

$$\tan(\sin \theta) - \sin \tan \theta = \frac{1}{30}\theta^7 + \frac{29}{756}\theta^9 + \&c.$$

Write in general

$$\begin{aligned} X &= \theta + A\theta^3 + B\theta^5 + C\theta^7 + D\theta^9 + \dots, \\ Y &= \theta + A'\theta^3 + B'\theta^5 + C'\theta^7 + D'\theta^9 + \dots \end{aligned}$$

Then, as far as θ^9 , we have

$$\begin{aligned} &X + A'X^3 + B'X^5 + C'X^7 + D'X^9 \\ &= \theta + A\theta^3 + B\theta^5 + C\theta^7 + D\theta^9 \\ &\quad + A'[\theta^3 + 3A\theta^5 + 3(A^2 + B)\theta^7 + (A^3 + 6AB + 3C)\theta^9] \\ &\quad + B'[\theta^5 + 5A\theta^7 + (10A^2 + 5B)\theta^9] \\ &\quad + C'[\theta^7 + 7A\theta^9] \\ &\quad + D'[\theta^9] \\ &= \theta \\ &\quad + \theta^3(A + A') \\ &\quad + \theta^5(B + 3AA' + B') \\ &\quad + \theta^7(C + 3A^2A' + 3A'B + 5AB' + C') \\ &\quad + \theta^9(D + 3A^3A' + 6AA'B + 3A'B + 6AB'A + 5BB' + 7AC' + D'). \end{aligned}$$

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and hence

$$\begin{aligned} &X + A'X^3 + B'X^5 + C'X^7 + D'X^9 \\ &\quad - Y - AY^3 - BY^5 - CY^7 - DY^9 \\ &= \theta^7[3AA'(A - A') + 2(AB' - A'B)] \\ &\quad + \theta^9[AA'(A^3 - A'^3) + 6AA'(B - B') + 4(AC' - A'C) + 10(A^2B'A - A'^2B)]. \end{aligned}$$

Now let

$$\begin{aligned} X = \sin \theta &= \theta - \frac{1}{6}\theta^3 + \frac{1}{120}\theta^5 - \frac{1}{5040}\theta^7 + \dots; \quad A = -\frac{1}{6}, \quad B = \frac{1}{120}, \quad C = -\frac{1}{5040}; \\ Y = \tan \theta &= \theta + \frac{1}{3}\theta^3 + \frac{2}{15}\theta^5 + \frac{17}{315}\theta^7 + \dots; \quad A' = \frac{1}{3}, \quad B' = \frac{2}{15}, \quad C' = \frac{17}{315}. \end{aligned}$$

we have therefore

$$\begin{aligned} AA' &= -\frac{1}{18}, \quad A - A' = -\frac{1}{2}, \quad A + A' = \frac{1}{6}, \quad B - B' = -\frac{1}{8}, \\ AB - A'B &= -\frac{1}{45}, \quad AC - A'C' = -\frac{1}{315}, \quad A^2B' - A'^2B = \frac{1}{216}. \end{aligned}$$

Hence

$$\begin{aligned} \text{coeff. } \theta^7 &= \frac{1}{12} - \frac{2}{45} = \frac{1}{30}, \\ \text{coeff. } \theta^9 &= \frac{1}{216} + \frac{1}{45} - \frac{1}{45} + \frac{1}{315} = \frac{29}{756}; \end{aligned}$$

and the required equation is thus verified.